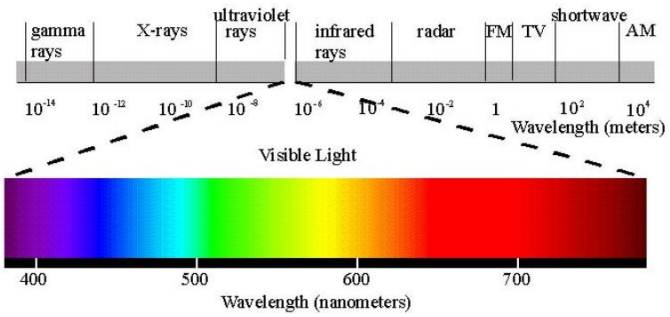
Lesson 11

Shading

- Shading is the process of determining the color of a pixel. i.e applying light (color) to the pixel.
- We understand that in electromagnetic spectrum, we have visible light (wavelength range from 390-700 nm)

Electromagnetic spectrum

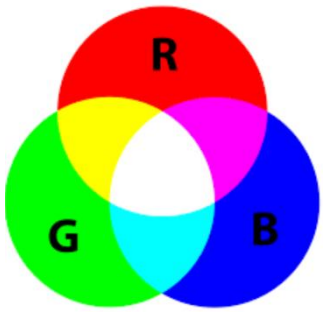


Color models

- RGB
- CMY(K)
- HSB

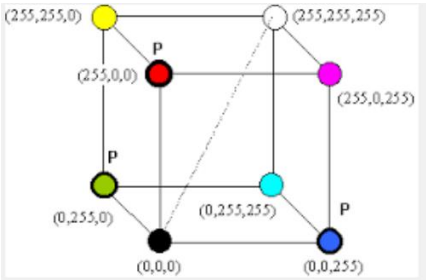
RGB model

In the RGB model each colour appears in its primary spectral components of red, green and blue



The model is based on a Cartesian coordinate system

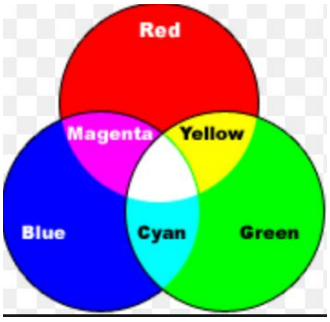
- RGB values are at 3 corners
- Cyan magenta and yellow are at three other corners
- Black is at the origin
- White is the corner furthest from the origin
- Different colours are points on or inside the cube represented by RGB vectors



CMY(K)

- Cyan, Magenta, Yellow (Black) – CMY(K)
- Known as subtractive color model

dye color	**absorbs**	**reflects**
cyan	red	blue and green
magenta	green	blue and red
yellow	blue	red and green
black	all	none



HSB (Hue, Saturation, Brightness)

HSB stands for Hue, Saturation and Brightness.

According to this model, any color is represented by 3 numbers.

The first number is the hue, and its value ranges from 0 to 360 degrees. Each degree represent a distinct color.

- First there is the red color (0 or 360 degrees)
- There are all other colors (for example
- yellow at 120 degrees,
- green at 180 degrees and
- blue at 240 degrees), up to the violet color.

All the rainbow's colors are represented here.

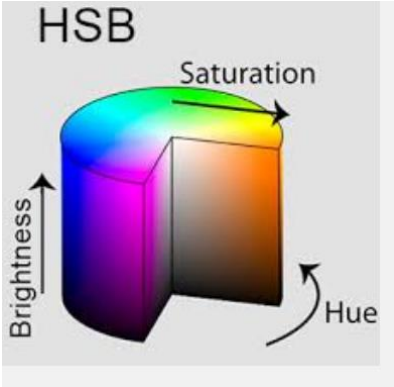
The second number is the saturation.

- It represents the amount of color or, more exactly, its percentage.
- Its value ranges from 0 to 100,
    - where 0 represents no color,
    - while 100 represents the full color.

Finally, the third number is the brightness.

- The color brightness can be enhanced by adding the white color, or you can reduce it adding the black color.

- In this case 0 represents the white color and 100 represents the black color.
- The more this value tends to 0, the brighter the color is. The more this value tends to 100 the darker the color is



Types of shading

There are three types of shading in polygonal rendering

- Flat (Per-polygon shading)
- Gourand (Per-vertex shading)
- Phong(Per-pixel shading)

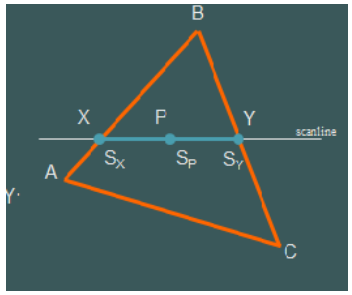
Flat shading

- Fast and simple.
- Compute the color of a polygon.
- Use that color on every pixel of the polygon.

Gouraud Shading

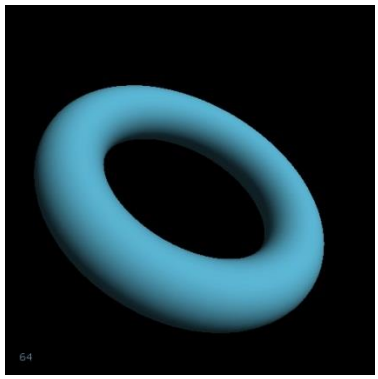
- Also fast and simple.
- Gives better sense of form than flat shading for many applications.
- Basic Idea:
    - Compute color at each vertex.

    - Bi-linearly interpolate(estimate value using two known values) color for each interior pixel.
- It is an interpolation method used to produce continuous shading of surfaces represented by polygon meshes.
- In practice, Gouraud shading is most often used to achieve continuous lighting on triangle surfaces by computing the lighting at the corners of each triangle and linearly interpolating the resulting colours for each pixel covered by the triangle.

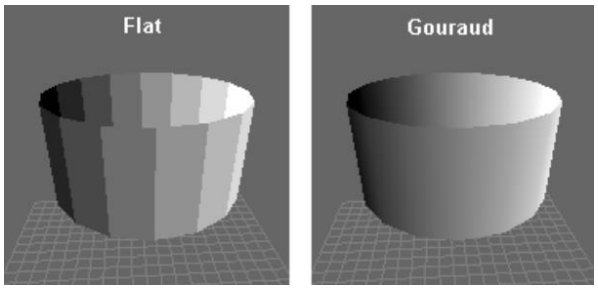


- Compute S_A, S_B, S_C for triangle ABC.
    - S_i = shade of point i.
- For a scanline XY, compute S_X, S_Y by lerping.(expanding)
    - e.g. $t_{AB} = |AX| / |AB|$.
    - $S_A = t_{AB} * S_A + (1-t_{AB}) * S_B$
- Compute S_P
    - By lerping between S_X and S_Y.

Gouraud output



Difference between Gourad and Flat shading



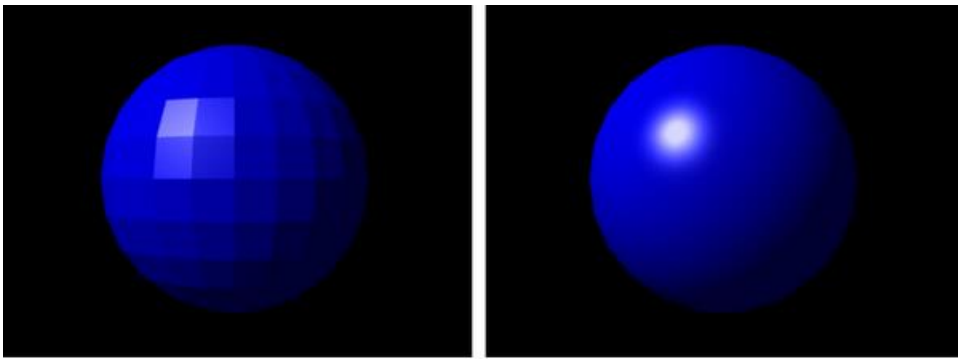
Phong Shading

- Phong shading applies lighting computation *per-pixel*.
- Uses linear interpolation of normal vectors, rather than colors.
- It improves upon Gouraud shading and provides a better approximation of the shading of a smooth surface.
- Phong shading assumes a smoothly varying surface normal vector.

Interpolate scan line endpoint normals N_a, N_b from endpoints of intercepted edges.

- Interpolate normal N_p at each pixel from N_a, N_b.
- Normalize N_p.

Difference between Flat and Phong shading

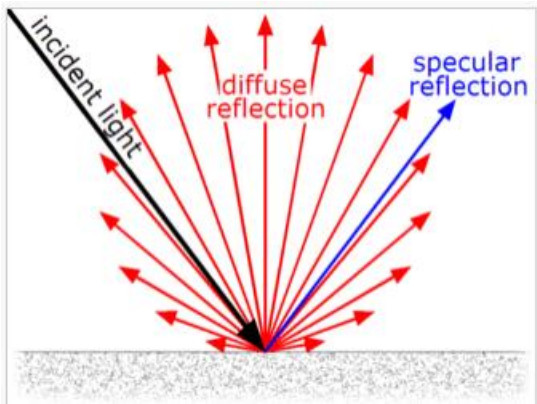


Phomg reflection model

Phong shading may also refer to the specific combination of Phong interpolation and the Phong reflection model, which is an empirical model of local illumination.

- Diffuse
- Ambient
- Specular

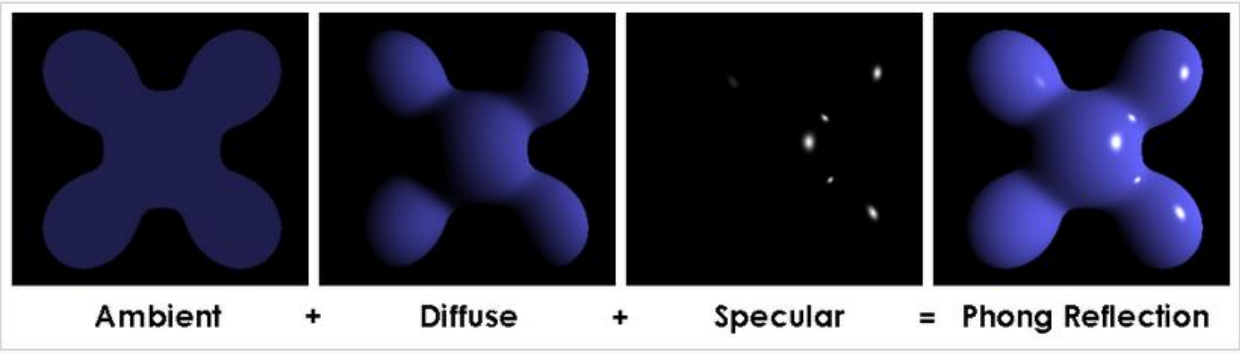
Diffuse reflection is the reflection of light from a surface (non-absorbing powder, paper or marble) such that an incident ray is reflected at many angles rather than at just one angle. The luminance is equal in all directions.



Specular reflection

Specular reflection is the mirror-like reflection of light (or of other kinds of wave) from a surface, in which light from a single incoming direction (a ray) is reflected into a single outgoing direction

The model describes the way a surface reflects light as a combination of the diffuse reflection of rough surfaces with the specular reflection of shiny surfaces



Other Types of Per-pixel Shading

Ray tracing.

- Doesn't use Gouraud or Phong shading.
- Each pixel uses own ray to determine color.
  - Can apply arbitrary lighting model.
  - Classical (Whitted) ray tracing uses Phong model.
- Since ray tracing determines colors based on intersections, don't have to use polygonal geometry.
  - Thus, can potentially use exact normals, rather than interpolation.